Chemically competent E. coli by the Inoue method

Version	Author	Effective	Review by
4.0	J. Smith	1 March 2026	1 March 2027

1. Purpose

This procedure produces ultra-competent *E. coli* DH5α cells with transformation efficiencies of 1–3 × 10^8 cfu/µg pUC19. The cells are grown at 18 °C, which is the key to the high efficiency, and are washed in Inoue transformation buffer before being snap-frozen in liquid nitrogen.

2. Materials and reagents

Reagent	Amount	Supplier · cat. no.
SOB medium	250 ml per prep, autoclaved and pre-warmed to 18 °C before inoculation	Prepared in-house
Inoue transformation buffer (ITB)	80 ml, ice-cold	see section 4
$MnCl_2 \cdot 4H_2O$	10.88 g/l	Sigma M3634
DMSO	1.5 ml per 20 ml of cells	Sigma D2650

3. Procedure

3.1 Growing the cells

1. Streak DH5α from a glycerol stock onto an LB plate and incubate overnight at 37 °C.
2. Pick a single colony into 25 ml SOB and grow for 6–8 h at 37 °C with shaking at 250 rpm.
3. Use this starter culture to inoculate three 1-litre flasks of SOB with 10, 4 and 2 ml respectively. Grow overnight at 18 °C with moderate shaking until one of the cultures reaches an OD600 of 0.55.

    1. Measure every 45 min once the OD600 passes 0.3.
    2. Discard the other two flasks.

3.2 Washing and freezing

4. Transfer the flask to an ice-water bath for 10 min.
5. Harvest the cells by centrifugation at 2,500 × g for 10 min at 4 °C.
6. Resuspend gently in 80 ml ice-cold ITB, centrifuge again and resuspend in 20 ml ITB.
7. Add 1.5 ml DMSO, mix by swirling and leave on ice for 10 min.
8. Dispense 50 µl aliquots into chilled tubes and snap-freeze in liquid nitrogen. Store at −80 °C.

Note: work quickly and keep everything ice-cold from this point on. Pre-chill pipettes, tubes and the centrifuge rotor.



Figure 1: Five flasks of SOB in the 18 °C shaker.

4. Inoue transformation buffer

- 55 mM $MnCl_2$
- 15 mM $CaCl_2$
- 250 mM KCl
- 10 mM PIPES (0.5 M, pH 6.7)

WARNING

DMSO penetrates skin and carries dissolved chemicals with it. Wear nitrile gloves. Liquid nitrogen causes cold burns: use a face shield and cryo-gloves when snap-freezing.

Quality control 1. Transform 100 pg of pUC19 into one aliquot of each batch and plate on LB with ampicillin; the batch passes when the efficiency is at least 10^8 cfu/µg. Record the result in the batch log together with the date and the OD600 at harvest.

Quality control 2. Transform 100 pg of pUC19 into one aliquot of each batch and plate on LB with ampicillin; the batch passes when the efficiency is at least 10^8 cfu/µg. Record the result in the batch log together with the date and the OD600 at harvest.

Quality control 3. Transform 100 pg of pUC19 into one aliquot of each batch and plate on LB with ampicillin; the batch passes when the efficiency is at least 10^8 cfu/µg. Record the result in the batch log together with the date and the OD600 at harvest.

Quality control 4. Transform 100 pg of pUC19 into one aliquot of each batch and plate on LB with ampicillin; the batch passes when the efficiency is at least 10^8 cfu/µg. Record the result in the batch log together with the date and the OD600 at harvest.

Quality control 5. Transform 100 pg of pUC19 into one aliquot of each batch and plate on LB with ampicillin; the batch passes when the efficiency is at least 10^8 cfu/µg. Record the result in the batch log together with the date and the OD600 at harvest.

Quality control 6. Transform 100 pg of pUC19 into one aliquot of each batch and plate on LB with ampicillin; the batch passes when the efficiency is at least 10^8 cfu/µg. Record the result in the batch log together with the date and the OD600 at harvest.

Quality control 7. Transform 100 pg of pUC19 into one aliquot of each batch and plate on LB with ampicillin; the batch passes when the efficiency is at least 10^8 cfu/µg. Record the result in the batch log together with the date and the OD600 at harvest.

Quality control 8. Transform 100 pg of pUC19 into one aliquot of each batch and plate on LB with ampicillin; the batch passes when the efficiency is at least 10^8 cfu/µg. Record the result in the batch log together with the date and the OD600 at harvest.

5. References

Inoue H, Nojima H, Okayama H (1990). High efficiency transformation of Escherichia coli with plasmids. Gene 96:23–28.